

Noninvasive Ventilation (BiLEVEL)

Noninvasive Ventilation assists the patient with pressure support, ventilation, and oxygenation without an invasive procedure such as intubation.

Indications for noninvasive ventilation

- Significant respiratory distress

Contraindications

- Inability to maintain open airway
- Systolic < 90 mmHg
- Apnea
- Suspected pneumothorax or chest trauma
- Patients at risk of aspiration (nausea/vomiting, foreign body obstruction, etc)
- Anatomy that would interfere with proper fit of mask (trauma, tracheostomy, etc)

Relative Contraindication

- Patient compensating for acidosis (i.e. Kussmaul respirations in DKA) – Discuss with Medical Control Physician

Equipment needed

- Ventilator
- Ventilator circuit
- Appropriate fitting Bipap mask
- +/- nebulizer set up if indicated

Noninvasive Ventilation Procedure

1. Prepare equipment
 - a. Initial Settings: IPAP 16, EPAP 8, FiO₂ 100%, start with respiratory rate that matches patient's rate
2. Adjust ventilator mode to () and verify/adjust settings
3. Place facemask on the patient (without straps) and verbally coach the patient to breathe deeply and allow the ventilator to support their efforts.
4. Completely attach the head strap to both sides of the facemask ensuring a tight, but tolerable, seal
5. Administer in-line nebulizer treatments for COPD/Asthma
6. Once the patient is compliant with the procedure, slowly increase IPAP/EPAP as needed

Note

- SpO₂ is affected by FiO₂ and PEEP/EPAP
- ETCO₂ is affected by rate and tidal volume